# Chapter 1

# FLUTTER

This chapter first provides an introduction to structural wind engineering and extends it to flutter. Next, the flutter mechanism and methods to obtain flutter derivatives are discussed. Then the mathematical formulation for calculating the critical wind speed is provided. Finally, the multi-frequency method is introduced and its workflow is compared with the single-frequency method.

## 1.1 Structural Wind Engineering

Structural wind engineering can be viewed as a field in civil engineering studying how a structure will respond when exposed to wind. This is studied by combining the concepts of fluid mechanics and structural mechanics to understand the mechanisms of different responses and quantify the load due to it. The response of a structure to a wind load can be broadly classified into two types: along and across wind response. Along-wind responses are forces acting on the structure parallel to the wind direction. In across-wind response, forces act perpendicular to the direction of the wind. Since wind loads are time-varying, the response of the structure to the wind load is dynamic in nature. Therefore, analysis of structures subjected to wind needs to be differentiated between static and dynamic analysis. In static analysis, the response of the structure due to the mean wind load is analyzed. In dynamic analysis, the response of the structure due to the fluctuating component (time-varying) of the wind load is considered. The static analysis primarily involves along-wind response created due to the drag force exerted by wind on the structure. The major sources of dynamic wind loads are responses due to extraneously induced excitations, instability-induced excitations, and movement-induced excitations.

Buffeting is one of the main examples of extraneously induced excitations, which is an along-wind response caused by turbulence in the wind. This shall be understood as, if a graph between wind velocity (acting on the structure) vs time is plotted, then the mean of this wind velocity is the static component and the variance of this plot is the buffeting component. The calculation of the buffeting component is given in EN 1991-1-4.

Next is instability-induced excitations. Vortex induced vibration (VIV) is an example of instability-induced excitations, which is an across-wind response that occurs in a bluff body. This is caused by the alternate shedding of vortices in the wake region of the structure. A bluff body shall be understood as an object that disturbs the streamline flow of a fluid and causes separation points. The wake is the region in the downstream of the body where the flow is highly disturbed and has negative pressures. When the fluid flows over a structure, the flow separates at the separation points and forms a wake region in the downstream. When the velocity (Reynolds number) of the fluid is low, the vortex is time-invariant and the structure is subjected to a constant along-wind force, that is, static and buffeting response. But if the velocity is high, the flow becomes unstable causing the vortices to shed alternately from the two sides of the structure. This induces an alternating time-varying force of a sinusoidal manner in the across-wind direction. This causes vibration in the structure known as vortex-induced vibration. Since this

is a dynamic force, it requires attention to the calculation of the frequency, as this can determine the resonance of the building. When the frequency of vortex shedding matches with the natural frequency of the structure, it will cause resonance leading to large amplitude vibrations and potential structural failure. The vortex shedding frequency is given by the formula f = St*U/D, where D is the characteristic length of the structure and U is the velocity of the fluid. St is the non dimensional parameter called Strouhal number, which is a shape constant. For regular shapes, the value of St shall be taken from EN 1991-1-4.

Finally, movement induced excitations are caused due to the interaction between the wind and the structure. When wind passes over a structure, it creates a force that causes deformation/displacement in the structure. Due to this displacement, the flow around the structure changes and consequently the force acting on the structure changes (also depending on the rate of change of displacement of the structure, the force acting changes). This can create a positive feedback loop where the deformation causes an increase in the force, which in turn causes more deformation. This can lead to a rapid increase in the amplitude of vibrations, which can ultimately result in structural failure. These phenomena are known as movement-induced vibrations, as an initial motion is first required to trigger this phenomenon.

Movement-induced excitations mostly cause instability failures in civil engineering structures. This is because once the threshold limit is reached, the structure would not be able to return to a stable state and would directly fail. Flutter and galloping are examples of these types of dynamic instabilities. Torsional divergence is also this type of instability, except this instability is static in nature. Torsional divergence can be observed in failure of solar panels, etc. Instability indicates a loss in any one of the structural parameters such as stiffness, damping, etc., due to the interaction with wind. In torsional divergence, the stiffness of the structure is lost due to the interaction with wind. This is analogous to the structural stability failure like buckling. In buckling, stiffness is lost due to the second-order effect, and in this case it is due to fluid-structure interaction (FSI). Also, similar to buckling, the failure due to this phenomenon depends on the flexibility of the structure rather than the ultimate strength [**?**].

In flutter, the loss is in damping. Since flutter is the focus of this thesis, the following sections will focus on discussing the mathematical formulations for quantifying this loss. Before discussing flutter, galloping is discussed first. This is done because galloping is a similar phenomenon to flutter with less complexity and hence can be useful in the intuitive understanding of these dynamic aeroelastic phenomena.

## 1.2 Galloping

Consider a 1-DOF spring-mass system with a bluff body attached to it. When the wind passes over the bluff body, it creates drag and lift forces. Assume, due to an external forced vibration or VIV, the structure starts to vibrate in the vertical direction. Due to this vertical vibration, depending on the structural velocity, the angle of attack of the wind on the bluff body changes. Since the angle of attack changes, the lift and drag forces also change (except for the sections where for any angle of attack the lift force remains constant - like circle). If the change in forces is such that it increases the motion

amplitude, then a positive feedback loop is created. This is known as galloping. For this reason, a purely circular rope or cable will not undergo galloping, as the lift force is constant for all angles of attack. But a rectangular section will undergo galloping as the lift force changes with angle of attack.

Next, the analytical formulation for galloping is derived. This formulation is based on the assumption: the instantaneous excitation force acting on the moving structure is nearly equivalent to the static force calculated considering change in angle of attack due to the velocity of the structure. This means there is no phase difference between applied force and structural velocity.

For the system shown in Figure 1.1(a), the net vertical force acting on the structure is

$$F_y = -F_L \cos\theta - F_D \sin\theta \tag{1.1}$$

where

$$F_L = C_L \frac{\rho d V_{\text{rel}}^2}{2}, \quad F_D = C_D \frac{\rho d V_{\text{rel}}^2}{2}.$$

$C_L$ is coefficient of lift force, $C_D$ is coefficient of Drag force and $\rho_{\text{air}}$ is the density of air.

$$V_{\text{rel}} = \sqrt{V^2 + \dot{y}^2}, \quad \theta = \tan^{-1}\left(\frac{\dot{y}}{V}\right)$$

- relative velocity of wind to structre's velocity

Writing the vertical force in terms of $C_y$,

$$F_y = C_y \frac{\rho d V_{\text{rel}}^2}{2}$$

where

$$C_y = -\frac{1}{\cos\theta}(C_L + C_D \tan\theta)$$

The equations of motion for the system can be written as

$$m\,\ddot{y} + c\,\dot{y} + k\,y = F_{\text{y}}(\alpha) = \frac{\rho_{\text{air}}\,V^2}{2}\,d\,C_y(\alpha) \tag{1.2}$$

To solve the equation, the system is linearised by assuming small motion at the zero angle of attack

$$\alpha \approx \frac{\dot{y}}{V} \tag{1.3}$$

For this condition

$$F_{\text{y}} \approx \left.\frac{\mathrm{d}F_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \alpha \tag{1.4}$$

which gives us the following equations

$$m\,\ddot{y} + c\,\dot{y} + k\,y = \left.\frac{\mathrm{d}C_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \frac{\dot{y}}{V}\frac{\rho\,V^2}{2}\,d \tag{1.5}$$

$$m\,\ddot{y} + \left(c\ -\ \left.\frac{\mathrm{d}C_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \frac{\rho\,V}{2}\,d\right)\dot{y} + k\,y = 0 \tag{1.6}$$

From this equation, it shall be realised how the system damping is affected by the interaction with wind: Effective damping $= \left(c\ -\ \left.\frac{\mathrm{d}C_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \frac{\rho V}{2}\,d\right)$. When this effective damping becomes negative, the system becomes unstable and galloping occurs. Hence, the critical wind speed for galloping can be obtained by setting the effective damping to zero, which yields,

$$V_{\text{crit}} = \frac{2c}{\left.\frac{\mathrm{d}C_y}{\mathrm{d}\alpha}\right|_{\alpha=0} \rho\,d} \tag{1.7}$$

To determine whether a structure will undergo galloping, CFD simulations are conducted without fluid-structure interaction (FSI) at various angles of attack. The lift and drag coefficients are extracted from these simulations. Subsequently, the slope of the aerodynamic force function at zero angle of attack is determined and substituted into the critical galloping equation presented above. This approach enables analysis of structural response subjected to unsteady forces using quasi-steady analysis. This is the result of the assumption introduced earlier in the galloping formulation.

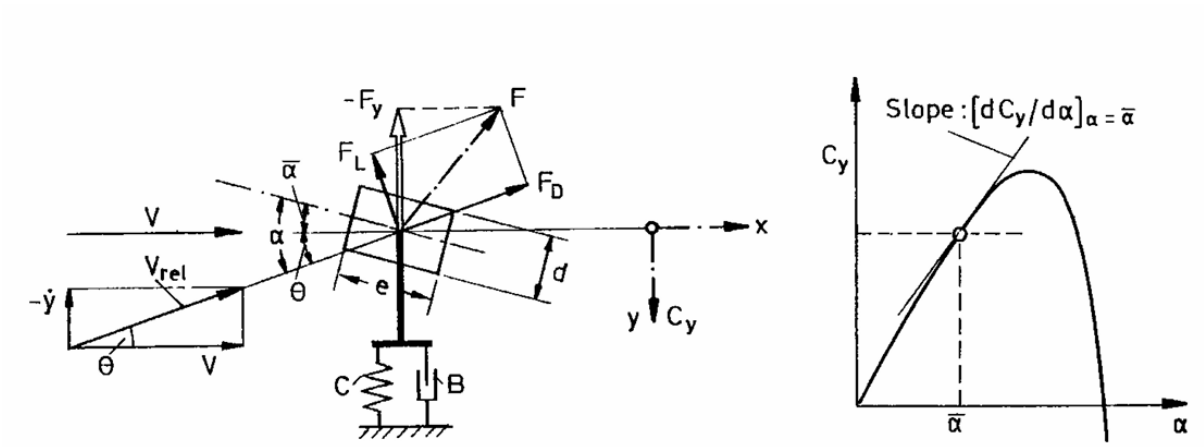



Figure 1.1: SDOF system for galloping and Cy Vs Angle of attack

## 1.3 Flutter Analysis

Flutter is a dynamic instability phenomenon that occurs when the aerodynamic forces acting on a structure couple with its natural modes of vibration, leading to self-excited oscillations. Consider a 2-DOF system of vertical (heave) and rotational (pitch) motion( Figure 1.2). If the vertical and pitch motion are coupled in such a way that the aerodynamic forces due to the vertical motion cause an increase in the pitch motion, and the

aerodynamic forces due to the pitch motion cause an increase in the vertical motion, then a positive feedback loop is created. This can lead to a rapid increase in the amplitude of oscillations, which can ultimately result in structural failure.

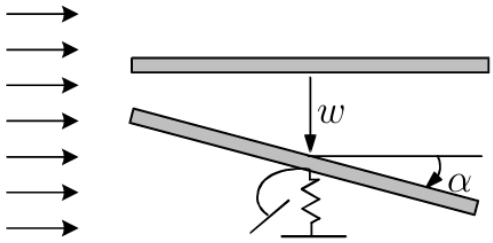



Figure 1.2: Two degree of freedom system for flutter analysis.

This is illustrated in Figure 1.3. In the first subfigure of 1.3, it shall be observed that the phase between the vertical and pitch motion is such that the positive work created during the upward motion is cancelled by the negative work created during the downward motion. Hence, the system is stable. But in the second image, the phase between the vertical and pitch motion is such that the positive work is created during both the upward and downward motion. Hence, a positive feedback loop is created and the system becomes unstable.

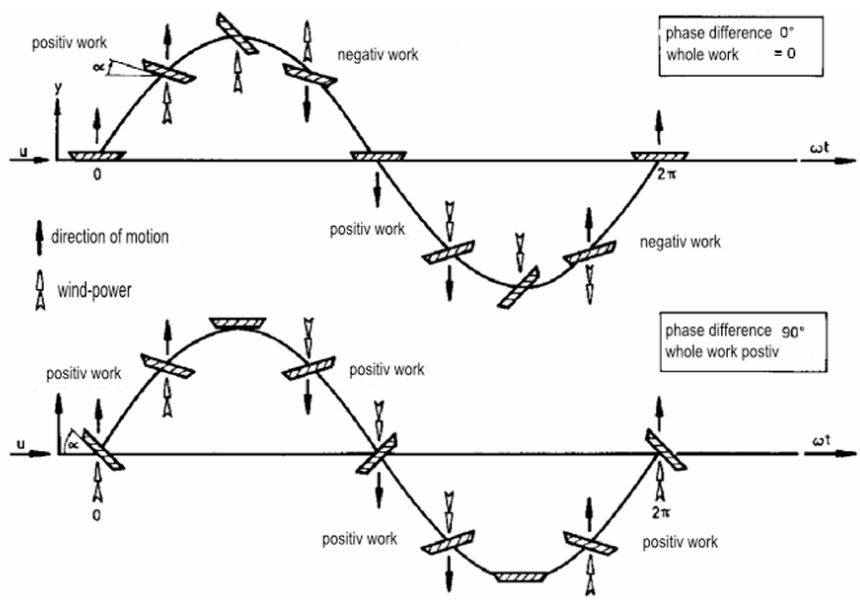



Figure 1.3: Flutter mechanism

Flutter analysis, like galloping, involves determining the critical wind speed at which structural instability initiates. However, unlike galloping (which is a single-degree-of-freedom system), flutter analysis considers two degrees of freedom and includes phase differences between the applied aerodynamic forces and the structural response. Consequently, flutter analysis must account for unsteady aerodynamic forces, making it considerably more complex than galloping.

The dynamic equilibrium equation for a 2-DOF system is

$$m(\ddot{h} + 2\omega_h \xi_h \dot{h} + \omega_h^2 h) = L_{ae} \tag{1.8}$$

$$I_\alpha(\ddot{\alpha} + 2\omega_\alpha \xi_\alpha \dot{\alpha} + \omega_\alpha^2 \alpha) = M_{ae} \tag{1.9}$$

$L_{ae}$ is the aerodynamic lift force due to wind and $M_{ae}$ aerodynamic moment due to wind. In this equation, quantifying unsteady aerodynamic forces is the challenge. In 1934, Theodorsen, a Norwegian American aerodynamicist, developed an analytical formulation for the aerodynamic forces for unsteady flutter analysis of an airfoil [**?**]. The solutions for $L_{ae}$ and $M_{ae}$ are functions of Theodorsen circulatory functions, which, in turn, are functions of $K$, the reduced frequency, where $K = \frac{\omega b}{U}$. $\omega b$ is oscillating frequency in radians per second, b is breadth of the cross-section and U is the velocity of the fluid. The reduced frequency shall be understood as, oscillating velocity in relative to fluid flow velocity. This quantifies the time lag between the applied force and the structure's displacement. As this theory was based on the assumption of potential flow, it cannot be applied to bridges. In 1971, Scanlan, an American civil engineer, reformulated Theodorsen's theory to accommodate bridges using a semi-empirical method based on eight functions (for a 2-DOF system) known as flutter derivatives [**?**].

$$L_{ae} = \frac{1}{2}\rho U^2 B \left[ K H_1^* \frac{\dot{h}}{U} + K H_2^* \frac{B\dot{\alpha}}{U} + K^2 H_3^* \alpha + K^2 H_4^* \frac{h}{B} \right] \tag{1.10}$$

$$M_{ae} = \frac{1}{2}\rho U^2 B^2 \left[ K A_1^* \frac{\dot{h}}{U} + K A_2^* \frac{B\dot{\alpha}}{U} + K^2 A_3^* \alpha + K^2 A_4^* \frac{h}{B} \right] \tag{1.11}$$

In this equation, $h$ denotes the vertical degree of freedom (called as heave motion) and $\alpha$ denotes rotational degree of freedom (pitch motion). The quantities $H_1^*$, $H_2^*$, $H_3^*$, $H_4^*$, $A_1^*$, $A_2^*$, $A_3^*$, $A_4^*$ are the flutter derivatives. These are shape constants that quantify the wind force acting on a structure as a linear relation of the section's vertical velocity, angular rotation, and speed of angular rotation. They are in turn functions of $U_{red} = \frac{1}{K} = \frac{U}{fB}$. These derivatives shall be viewed as: how $C_d$ acts as a non-dimensional shape constant for computing drag force on a structure, each of the flutter derivatives is for quantifying the aeroelastic lift and moment forces. For example, $H_1^*$ is the flutter derivative for quantifying the lift force due to vertical velocity of the structure. $A3^*$ is the flutter derivative for quantifying the moment force due to angular rotation of the structure.

Since this is a semi-empirical method, these derivatives must be found through CFD simulations or wind-tunnel tests. Once obtained, they are applied back to the dynamic equilibrium equation to determine the critical velocity. The approaches for finding the flutter derivatives can be broadly classified based on the nature of the applied motion: forced motion tests or free vibration. In forced motion tests, the structure is forced to oscillate at a specific frequency and amplitude, and the resulting aerodynamic forces are measured to determine the flutter derivatives. In free vibration tests, the structure is allowed to vibrate freely under the influence of aerodynamic forces, and the flutter derivatives are determined from the observed damping response. For the preliminary analysis of flutter, forced motion tests with CFD have greater importance due to their

easier setup and lower cost. Since forced motion tests are the interest of this thesis, only it is explained in the following sections

## 1.4 Identification of Flutter Derivatives with Forced Motion Simulations

A 2-D cross-section of the bridge deck is modelled in a virtual wind tunnel with appropriate boundary conditions. The bridge cross-section is then oscillated sinusoidally in the vertical (heave) direction according to

$$h(t) = h_0 \sin(\omega t), \tag{1.12}$$

$$\dot{h}(t) = \omega h_0 \cos(\omega t), \tag{1.13}$$

where $h_0$ is the maximum amplitude of the applied heave motion and $\omega$ is the angular frequency of the applied heave motion. Since no pitch motion is applied, $\alpha = \dot{\alpha} = 0$, and the aerodynamic lift force1.10 and pitching moment1.11 simplifies to

$$L_{ae} = \frac{1}{2}\rho U^2 B \left[ K H_1^* \frac{\dot{h}}{U} + K^2 H_4^* \frac{h}{B} \right] \qquad \text{(Scanlan formulation)}, \tag{1.14}$$

$$M_{ae} = \frac{1}{2}\rho U^2 B^2 \left[ K A_1^* \frac{\dot{h}}{U} + K^2 A_4^* \frac{h}{B} \right] \qquad \text{(Scanlan formulation)}. \tag{1.15}$$

Under the small-amplitude assumption (small $h_0$), Scanlan's linearisation implies that the aerodynamic lift and moment measured from CFD take the sinusoidal form

$$L_{ae} = B_0 \sin(\omega t + \phi) \;=\; B_1 \cos(\omega t) + B_2 \sin(\omega t), \tag{1.16}$$

$$M_{ae} = C_0 \sin(\omega t + \Phi) \;=\; C_1 \cos(\omega t) + C_2 \sin(\omega t). \tag{1.17}$$

The flutter derivatives are determined by equating the lift force from the Scanlan formulation with the CFD result and matching cosine and sine coefficients, which gives

$$\frac{1}{2}\rho U^2 B \left[ K H_1^* + K^2 H_4^* \right] = B_1 \cos(\omega t) + B_2 \sin(\omega t), \tag{1.18}$$

$$\frac{1}{2}\rho U^2 B^2 \left[ K A_1^* + K^2 A_4^* \right] = C_1 \cos(\omega t) + C_2 \sin(\omega t). \tag{1.19}$$

From this, the flutter derivatives for heave excitation are obtained as

$$H_1^* = \frac{2B_1}{\rho\,\omega^2 B^2 h_0}, \qquad H_4^* = \frac{2B_2}{\rho\,\omega^2 B^2 h_0}, \tag{1.20}$$

$$A_1^* = \frac{2C_1}{\rho\,\omega^2 B^3 h_0}, \qquad A_4^* = \frac{2C_2}{\rho\,\omega^2 B^3 h_0}. \tag{1.21}$$

The same procedure is repeated with pitch (torsional) excitation to obtain the $H_2^*$, $H_3^*$, $A_2^*$, and $A_3^*$ derivatives.

$$H_2 = \frac{2B_1}{\rho\omega^2 B^3\theta_0}, \qquad H_3 = \frac{2B_2}{\rho\omega^2 B^3\theta_0}, \tag{1.22}$$

$$A_2 = \frac{2C_1}{\rho\omega^2 B^4\theta_0}, \qquad A_3 = \frac{2C_2}{\rho\omega^2 B^4\theta_0}. \tag{1.23}$$

Consequently, a total of two forced-motion simulations (one heave, one pitch) are required to obtain a complete set of eight flutter derivatives at each reduced velocity.

This is the single frequency method where a heave/pitch motion of one frequency is applied to the structure to find the flutter derivative. Obtaining flutter derivatives with this approach is highly computationally costly because of the following reasons. Since flutter derivatives are functions of reduced velocity, multiple CFD simulations (by varying the frequency of input forced motion) must be performed to obtain them. For finding each point in a set of 8 flutter derivatives, 2 CFD simulations are required. To obtain an accurate representation of each flutter derivative function, a minimum of 6 to 8 points on the flutter derivative curve is needed. Hence, a total of 16 simulations are required. Each CFD simulation will take approximately 48 hours. Hence, to obtain a complete set of flutter derivatives, 768 hours of simulation time are required.

## 1.5 Multi-Frequency Method

In the multi-frequency method, instead of applying the single-frequency forced-motion oscillation at a time and extracting the 4 flutter derivatives, several frequencies are superimposed and applied simultaneously to the structure. By this way, N*4 flutter derivatives can be obtained from N simulations. For example, if 4 frequencies are applied simultaneously, then 16 flutter derivatives can be obtained from a single simulation. The superimposition of multiple frequencies can be expressed as

$$h(t) = \sum_{k=1}^{N} h_{0,k} \sin(\omega_k t + \phi_k). \tag{1.24}$$

Under the assumption of aerodynamic linearity, the total aerodynamic response is the superposition of the responses at each individual frequency. Consequently, $N$ sets of flutter derivatives can be extracted from a single simulation by applying a Fourier decomposition of the measured aerodynamic forces at each frequency component $\omega_k$.

The procedures for calculating the flutter derivatives (Section 2) and the critical wind speed (Section 3) are identical for both the single- and multi-frequency methods. The only difference lies in the identification of aerodynamic forces (Section 1), where multiple frequency components are excited simultaneously rather than one at a time. Figures 1.4 and 1.5 illustrate the workflow for the two methods.

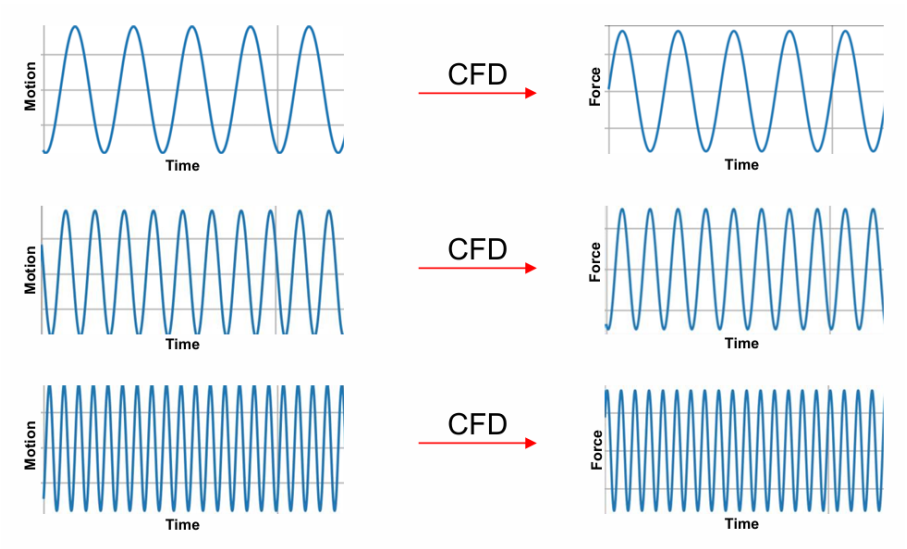



Figure 1.4: Workflow for the single-frequency method.

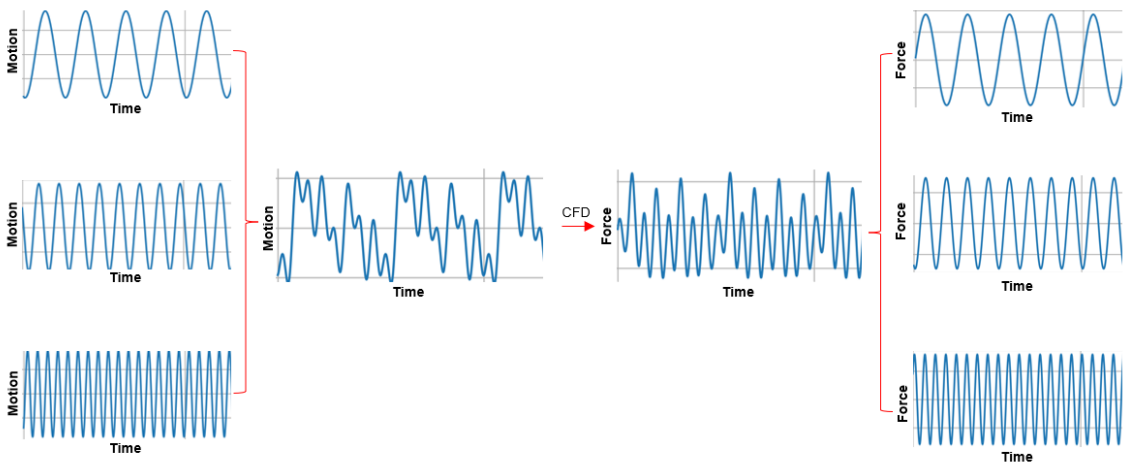



Figure 1.5: Workflow for the multi-frequency method.

## 1.6 Calculation of Critical Wind Speed

To obtain the critical wind speed, the equations of motion for the bridge deck (equations 1.8 and 1.9) are equated to the aerodynamic forces from the Scanlan formulation (1.10 and 1.11) and solved. On equating it yields

$$m\Big(\ddot{h} + 2\omega_h\zeta_h\dot{h} + \omega_h^2 h\Big) = \frac{1}{2}\rho U^2 B\left[KH_1^*\frac{\dot{h}}{U} + KH_2^*\frac{B\dot{\alpha}}{U} + K^2H_3^*\alpha + K^2H_4^*\frac{h}{B}\right], \quad (1.25)$$

$$I_\alpha\big(\ddot{\alpha} + 2\omega_\alpha\zeta_\alpha\dot{\alpha} + \omega_\alpha^2\alpha\big) = \frac{1}{2}\rho U^2 B^2\left[KA_1^*\frac{\dot{h}}{U} + KA_2^*\frac{B\dot{\alpha}}{U} + K^2A_3^*\alpha + K^2A_4^*\frac{h}{B}\right]. \quad (1.26)$$

Converting this to matrix format and rearranging the terms

$$\begin{aligned}&\begin{bmatrix} m & 0 \\ 0 & I \end{bmatrix}\begin{Bmatrix} \ddot{h} \\ \ddot{\alpha} \end{Bmatrix} + \begin{bmatrix} C_h & 0 \\ 0 & C_\alpha \end{bmatrix}\begin{Bmatrix} \dot{h} \\ \dot{\alpha} \end{Bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\alpha \end{bmatrix}\begin{Bmatrix} h \\ \alpha \end{Bmatrix} \\ &- \frac{1}{2}\rho B^2\left(\begin{bmatrix} H_1^* & BH_2^* \\ BA_1^* & B^2A_2^* \end{bmatrix}\begin{Bmatrix} \dot{h} \\ \dot{\alpha} \end{Bmatrix} + \begin{bmatrix} H_4^* & BH_3^* \\ BA_4^* & B^2A_3^* \end{bmatrix}\begin{Bmatrix} h \\ \alpha \end{Bmatrix}\right) = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}.\end{aligned} \quad (1.27)$$

This equation is in time domain. To solve it, assuming a harmonic motion $h = e^{i\omega t}$ and $\alpha = e^{i\omega t}$ and transforming it to the frequency domain. The system becomes

$$\left(-\omega^2\begin{bmatrix} m & 0 \\ 0 & I \end{bmatrix} + i\omega\begin{bmatrix} C_h & 0 \\ 0 & C_\alpha \end{bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\alpha \end{bmatrix} - \frac{1}{2}\rho B^2\begin{bmatrix} H_4^* - H_1^* & BH_3^* - BH_2^* \\ BA_4^* - BA_1^* & B^2A_3^* - B^2A_2^* \end{bmatrix}\right)\begin{Bmatrix} h \\ \alpha \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \quad (1.28)$$

which is of the standard form

$$-\omega^2 M + i\omega C + K = 0. \quad (1.29)$$

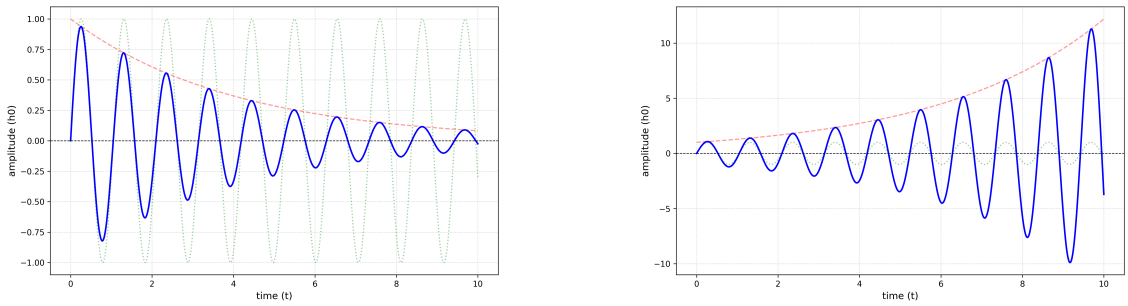
The unknown in the equation is $\omega = \omega_r + i\omega_i$, which represents the frequency in complex form: The meaning of frequency in complex form can be interpreted from structural dynamics theory. Solution of a simple harmonic oscillation without damping can be represented with $x(t) = e^{i\omega t} = \cos(\omega t) + \mathrm{i}\,\sin(\omega t)$. This represents a pure sinusoidal oscillation with frequency $\omega$ (real number) and with phase angle. To represent the effect of damping, the solution takes the form $x(t) = e^{i\omega t} = e^{i\omega_r t}\cdot e^{-\omega_i t} = e^{-\omega_i t}\big(\cos(\omega t) + i\sin(\omega t)\big)$, where the real part $\omega_r$ is the oscillation frequency and the imaginary part $\omega_i$ governs the damping of the response. This equation has to be solved for $\omega$ to determine the system's oscillating frequency and damping characteristics. Because structural damping is present, the problem takes the form of a quadratic eigenvalue problem. Hence this is linearised as shown below to be solved as a generalised eigenvalue problem:

$$\left(\begin{bmatrix} 0 & K \\ K & iC \end{bmatrix} - \omega\begin{bmatrix} K & 0 \\ 0 & -M \end{bmatrix}\right)\begin{pmatrix} X \\ \omega X \end{pmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \qquad \text{i.e.} \quad A - \lambda B = 0. \quad (1.30)$$

Based on the explanation provided above for damping, the system stability or instability can be determined. For a case, consider a solution of $\omega = \omega_r + i\omega_i$, where both are positive. Then, $e^{-\omega_i t}\big(\cos(\omega t) + i\sin(\omega t)\big)$ represents a damped oscillation with frequency $\omega_r$ and an amplitude that decays exponentially over time due to the positive $\omega_i$. If $\omega = \omega_r - i\omega_i$, then $e^{+\omega_i t}\big(\cos(\omega t) + i\sin(\omega t)\big)$ represents the same oscillation but with an amplitude that grows exponentially over time due to the negative $\omega_i$. So, the sign of $\omega_i$ determines stability:

- $\omega_i > 0$: $\quad h = h_0\, e^{i\omega_r t} \cdot e^{-\omega_i t}$ — stable (amplitude decays).
- $\omega_i < 0$: $\quad h = h_0\, e^{i\omega_r t} \cdot e^{+|\omega_i| t}$ — unstable (amplitude grows; flutter onset).

This is illustrated in Figure 1.6.




(a) Damped solution

(b) Diveregent solution

Figure 1.6: Time-domain response of the flutter/galloping solution

Determining the critical wind speed is an iterative process because the flutter derivatives are functions of reduced velocity, which in turn is a function of wind speed. Hence the following steps (which form the algorithm of the code) are performed iteratively to find the critical wind speed:

1. A reduced velocity $U_r = U/(fB)$ is selected.
2. Flutter derivatives corresponding to that reduced velocity are interpolated from the flutter-derivative functions.
3. The linearised eigenvalue problem is solved for $\omega$.
4. If the system is stable, the reduced velocity is increased and the process repeats until the system first becomes unstable. The critical wind speed is the speed at which $\omega_i$ changes sign from positive to negative, indicating the onset of flutter instability.
5. The actual critical wind speed is then recovered from

$$U_c = U_{r,c}\,\frac{\omega_r}{2\pi}\,B, \tag{1.31}$$

where $U_{r,c}$ is the reduced critical velocity and $\omega_r$ is the real part of the eigenfrequency at that point.

## 1.7 Validation of the Critical Velocity Tool

The Python tool is validated against the benchmark results of [**?**]. The structural paramters taken for the validation are taken from [**?**] given in Table 1.1

Table 1.1: Structural parameters used for validation (Diana benchmark).

| Symbol | Description | Value |
|---|---|---|
| $m_L$ | Mass per unit length | 22 740 kg/m |
| $J_L$ | Moment of inertia per unit length | $2.47 \times 10^6$ kg m$^2$/m |
| $B$ | Deck chord (width) | 31 m |
| $f_z$ | Vertical structural eigenfrequency | 0.10 Hz |
| $f_\theta$ | Torsional structural eigenfrequency | 0.278 Hz |
| $\xi$ | Damping ratio (both modes) | 0.003 (−) |

Theodorsen derivatives are used for the aerodynamic properties of the cross-section. The Theodorsen flutter derivatives are analytical functions derived from the closed-form solution for a thin flat plate in potential flow, first obtained by Theodorsen [**?**]. They are expressed in terms of the real and imaginary parts of the Theodorsen circulatory function $C(k) = F(k) + iG(k)$, where $k$ is the reduced frequency. The analytical expressions are listed in Table 1.2 and plotted in Figure **??**.

Table 1.2: Analytical Theodorsen flutter derivative expressions for a flat plate.

| | $H_i^*$ | $A_i^*$ |
|---|---|---|
| $i = 1$ | $H_1^* = -\dfrac{2\pi}{K} F$ | $A_1^* = \dfrac{\pi}{2K} F$ |
| $i = 2$ | $H_2^* = -\dfrac{\pi}{2K}\left(1 + F + \dfrac{4G}{K}\right)$ | $A_2^* = -\dfrac{\pi}{8K}\left(1 - F - \dfrac{4G}{K}\right)$ |
| $i = 3$ | $H_3^* = -\dfrac{2\pi}{K^2}\left(F - \dfrac{KG}{4}\right)$ | $A_3^* = \dfrac{\pi}{2K^2}\left(\dfrac{K^2}{32} + F - \dfrac{KG}{4}\right)$ |
| $i = 4$ | $H_4^* = \dfrac{\pi}{2}\left(1 + \dfrac{4G}{K}\right)$ | $A_4^* = -\dfrac{\pi}{2K} G$ |

### Results

The results obtained with the Python tool are shown in Figures 1.7 and **??**.

Since the bridge deck is modelled as a two-degree-of-freedom system, two eigenmodes are obtained. The zero-crossing of the imaginary part of each eigenfrequency is checked; the lowest reduced velocity at which a zero crossing occurs is taken as the reduced critical velocity $U_{r,c}$. Substituting into Equation (1.31) yields

$$\begin{aligned} U_{r,c} &= 12.9, \\ \omega_r &= 1.21 \text{ rad/s}, \\ U_c &= 77.32 \text{ m/s}. \end{aligned}$$

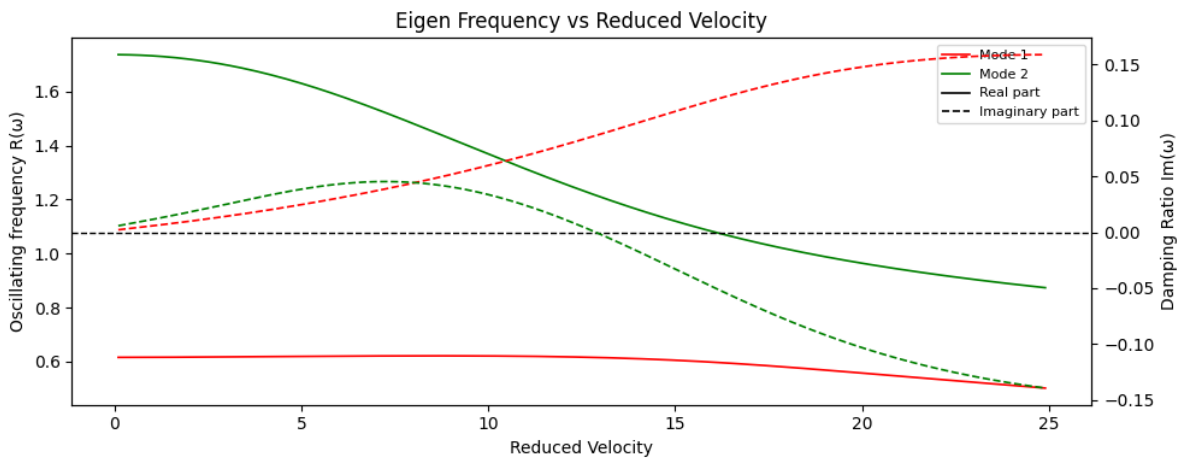



Figure 1.7: Real and imaginary parts of the eigenfrequency plotted against reduced velocity

This result falls within the reference range of [77.1, 78.6] m/s reported by [**?**], confirming the validity of the tool.